

Naden Mohamed Yasen

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Summary

Motivated Computer Science & AI student with hands-on experience in data science, ML, NLP, and LLMs & RAG. Skilled at building practical ML solutions and communicating insights clearly through visualizations and dashboards. A strong team player with leadership potential, solid communication skills, and a passion for continuous learning.

Education

BSc Computer Science and Artificial Intelligence, Helwan University

GPA: Very Good (3.37 / 4)

Projects

Healthcare Chatbot System with RAG

- Built a **Healthcare Chatbot with RAG** leveraging **Groq API** for accurate, context-aware medical Q&A.
- Implemented **multi-source retrieval** from Wikipedia API, DuckDuckGo (trusted sites), and multi-format document uploads (PDF, Word, PowerPoint, Text).
- Engineered **content preprocessing pipeline** (cleaning, chunking) with **HuggingFace embeddings (Sentence Transformer)** + **FAISS vector store** for efficient semantic search.
- Developed a **hybrid retrieval scoring mechanism** combining semantic similarity with credibility weighting.
- Designed a **Streamlit interface** with session-based state management (**LangGraph**) to handle multi-turn conversations seamlessly.

Egypt Tourism sentiment analysis

- Developed a comprehensive sentiment analysis pipeline to understand foreign tourists' opinions about tourism in Egypt by scraping reviews from **TripAdvisor** and **Quora**.
- Conducted extensive **text preprocessing**, handled class imbalance, Implemented and compared multiple ML models: **Logistic Regression**: 89.6% accuracy, **XGBoost**: 90.7% accuracy
- Built deep learning models **BiLSTM** and **GRU** (both achieved **91% accuracy, 88% F1 score**), using **TensorFlow**.
- Integrated Large Language Models (**Google Gemini Flash API**) with **Langchain** to generate actionable recommendations for visitors, triggered by negative sentiment detection to provide improvement suggestions.
- Visualized insights using **word clouds**, **confusion matrices**, **bar plots**, and an interactive **Dash dashboard** to present sentiment distribution and model performance, deployed models using **Streamlit**.

Twitter Sentiment Analysis using Machine Learning & LSTM

- Developed an end-to-end sentiment analysis pipeline using the Sentiment140 dataset (1.6M tweets).
- Applied extensive text preprocessing (tokenization, stemming, emoji normalization) and developed multiple classifiers (**logistic regression, SVM, XGBoost**), achieving **80% accuracy** with TF-IDF + LR.
- Engineered custom **Word2Vec** embeddings and trained an **LSTM**, achieving **77% accuracy** using **Keras** with dropout regularization and early stopping. Saved trained models for reuse using **Pickle**.

Professional Experience

AI & Data Science intern, Digital Egypt Pioneers Initiative DEPI, (November 2024 - May 2025)

Completed a comprehensive program provided by IBM, covering data science fundamentals, Python, data analysis, machine learning, SQL, and MLOps tools including MLflow and Hugging Face. Finally completed the hands-on capstone project.

Technical Skills

- + **Programming Languages:** Python, C, C++, Java,
- + **Libraries:** Pandas, NumPy, Scikit-learn, Matplotlib, Seaborn, Plotly, Dash, NLTK, Pytorch, Streamlit, FastAPI.
- + **Tools:** Jupyter Notebook, Git, GitHub, VSCode, Google Colab, Postman.
- + **Concepts:** Data Structure & Algorithms, Object-Oriented Programming (OOP), Database & SQL, Design Patterns, Solid Principles, Clean Architecture, Data Cleaning & Preprocessing, Data Visualization, Exploratory data analysis (EDA), Machine Learning, Deep Learning, Supervised/Unsupervised Learning, Text Preprocessing, (NLP), Sentiment Analysis, Model Evaluation, LLMs, RAG, AI Agents, Fine Tuning, Prompt Engineering..
- + **Languages:** Excellent knowledge of both spoken and written English and Arabic.